1ST WEEK OF DEVELOPMENT

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Ovulation to Implantation

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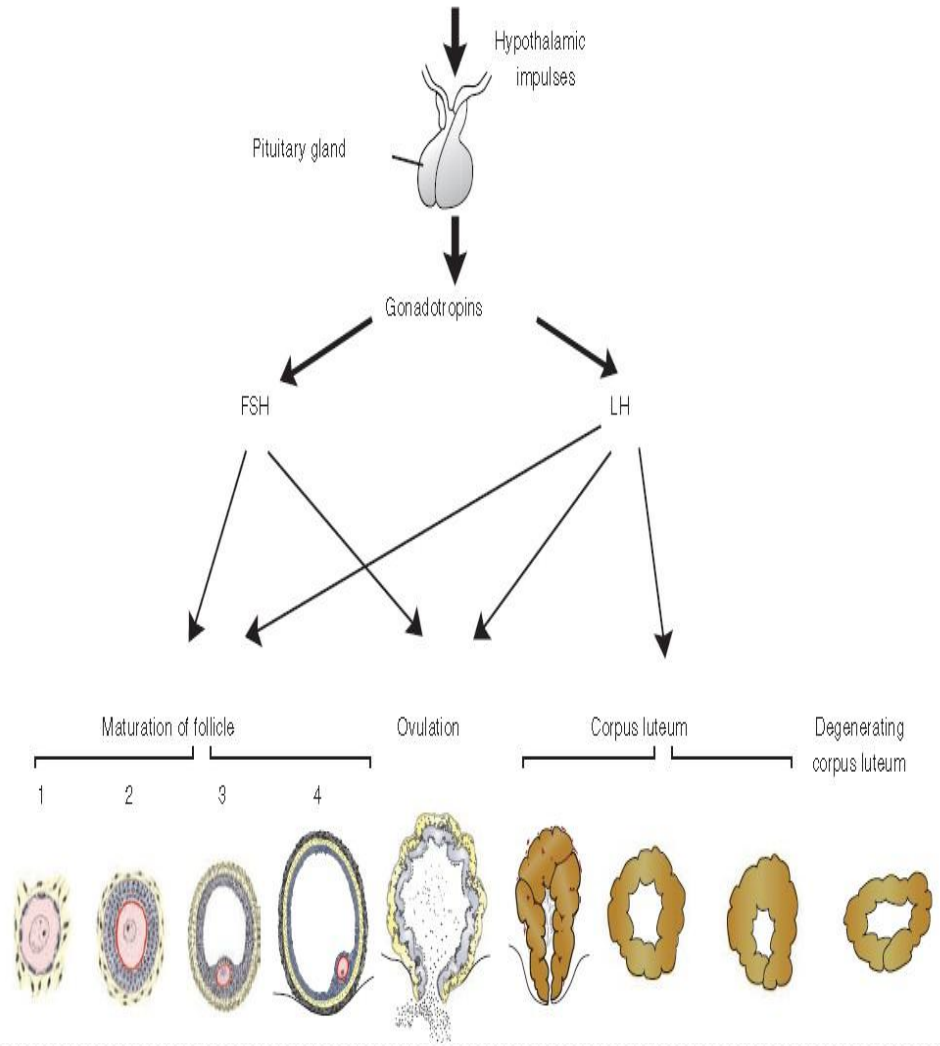
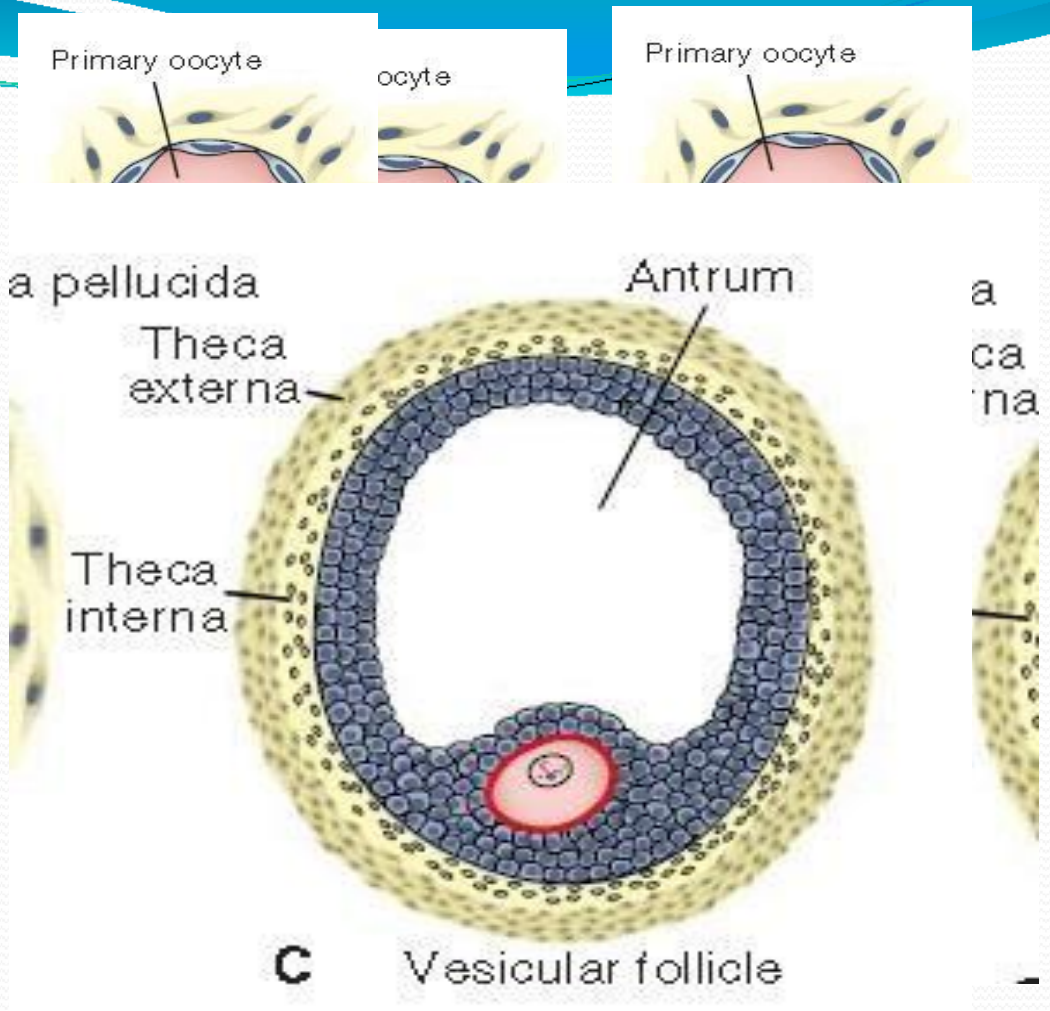
Ovarian Cycle:

- At puberty (GnRH) produced by the hypothalamus acts on the anterior pituitary gland to secrete:
- ✓ follicle-stimulating hormone (FSH)
- ✓ luteinizing hormone (LH),
- Stimulate and control cyclic changes in the ovary

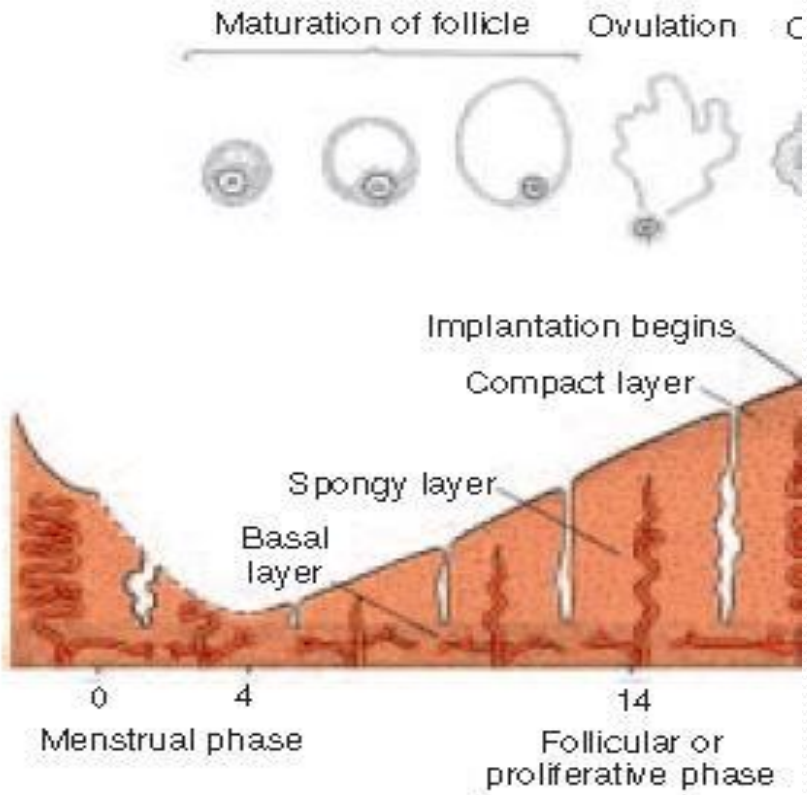


- Primary (preantral) stage follicles are stimulated to grow under the influence of FSH
- Only one of these follicles reaches full maturity, the others degenerate.
- FSH also stimulates maturation of follicular (granulosa) cells surrounding the oocyte



- **granulosa and thecal cells produce estrogens:**

1. **uterine endometrium enters the proliferative phase**
2. **thinning of the cervical mucus**
3. **stimulate the pituitary gland to secrete LH**

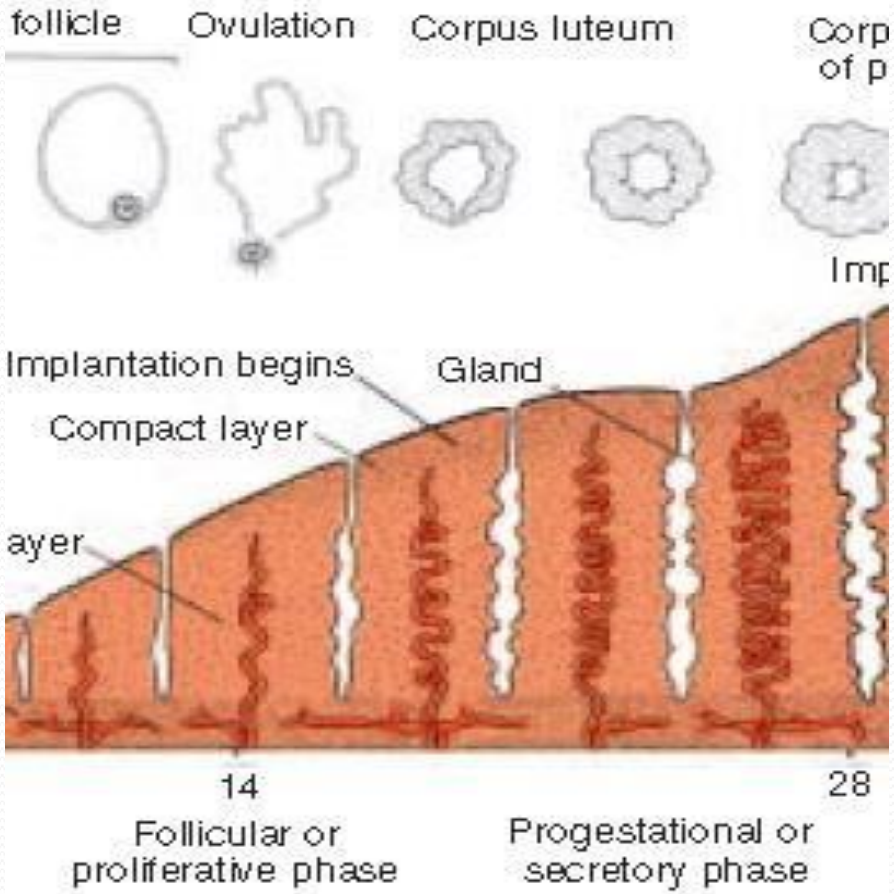


- At mid-cycle:
- LH surge that

a) causing oocytes to complete meiosis I and initiate meiosis II

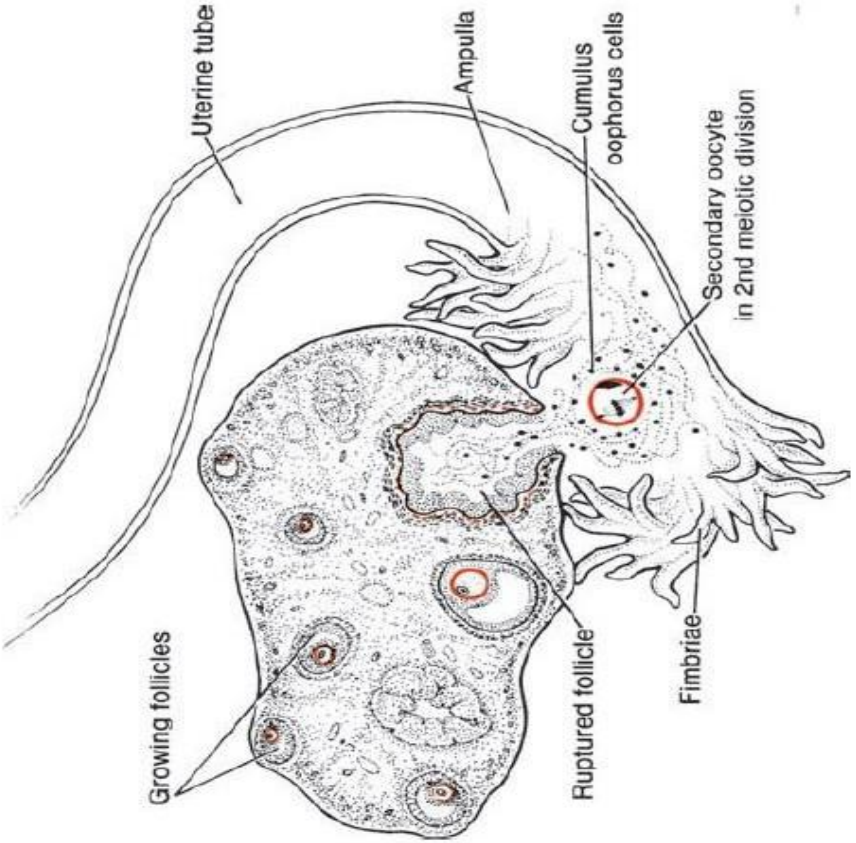
b) stimulates production of progesterone by follicular stromal cells

c) causes follicular rupture and ovulation.

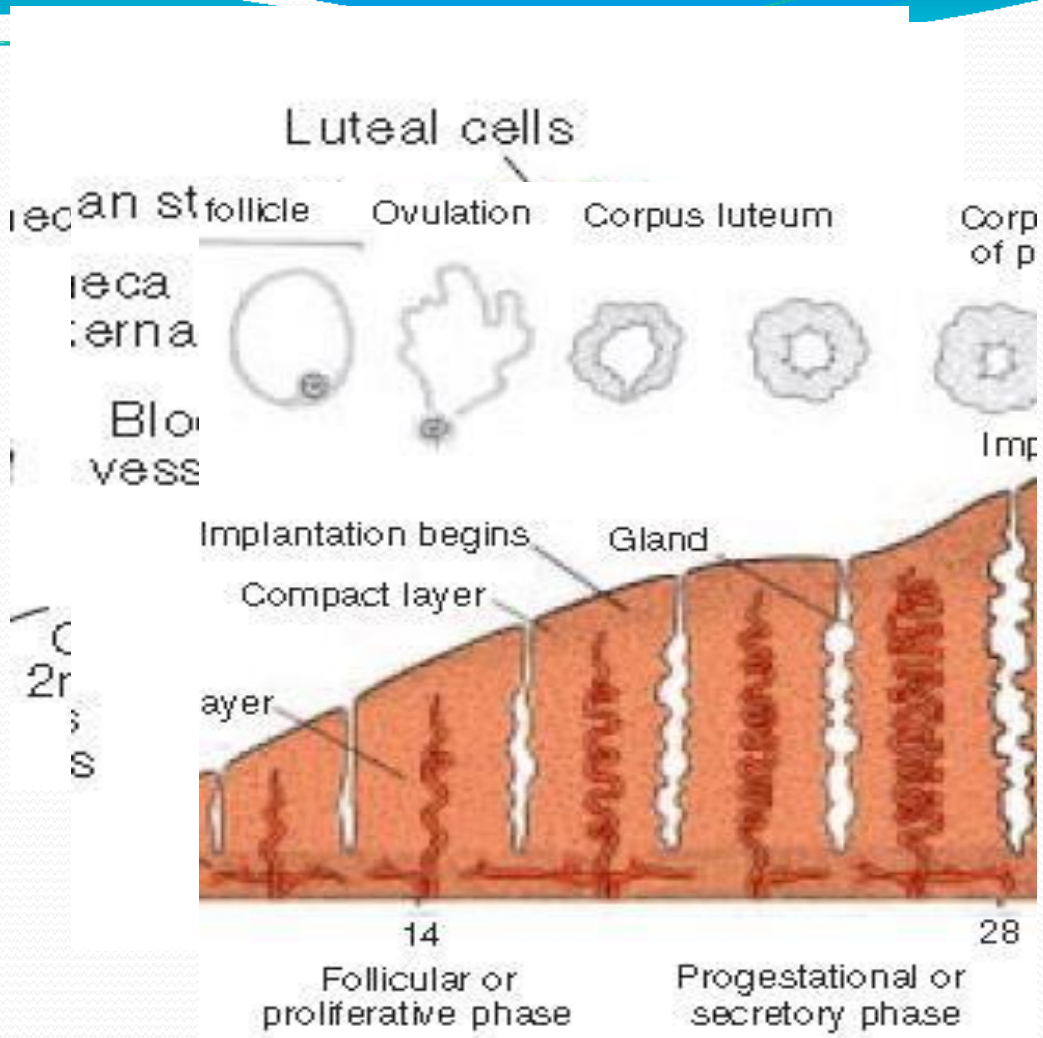


OVULATION:

- LH that causes the primary oocyte to complete meiosis I and the follicle to enter the preovulatory stage
- Meiosis II is also initiated
- Localized contractions extrude the oocyte with its surrounding granulosa

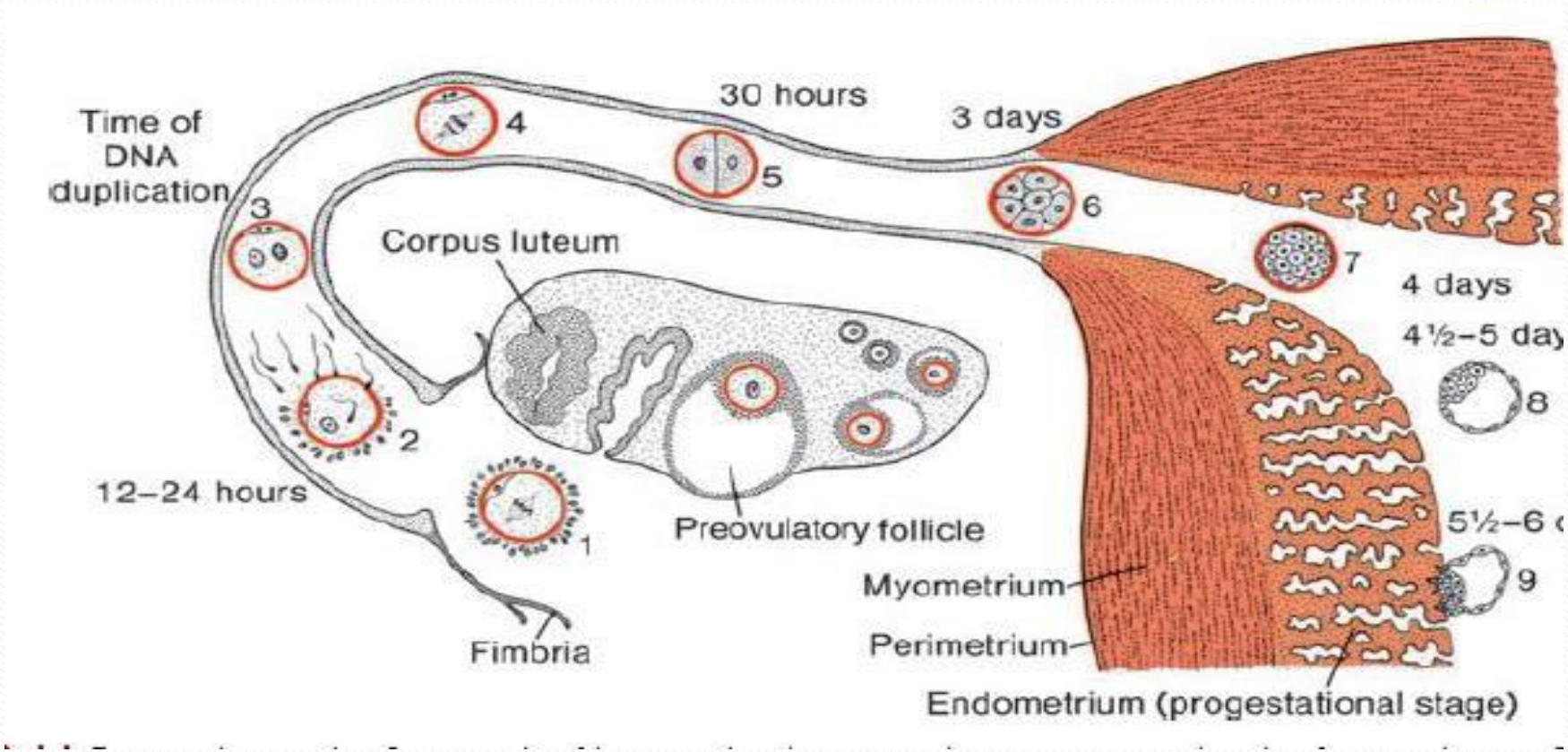


- **After ovulation:**
- **granulosa cells remaining in the ruptured follicle form (corpus luteum) which secrete the hormone progesterone:**
- **progesterone and estrogens stimulate uterine mucosa to enter the progestational or secretory phase**

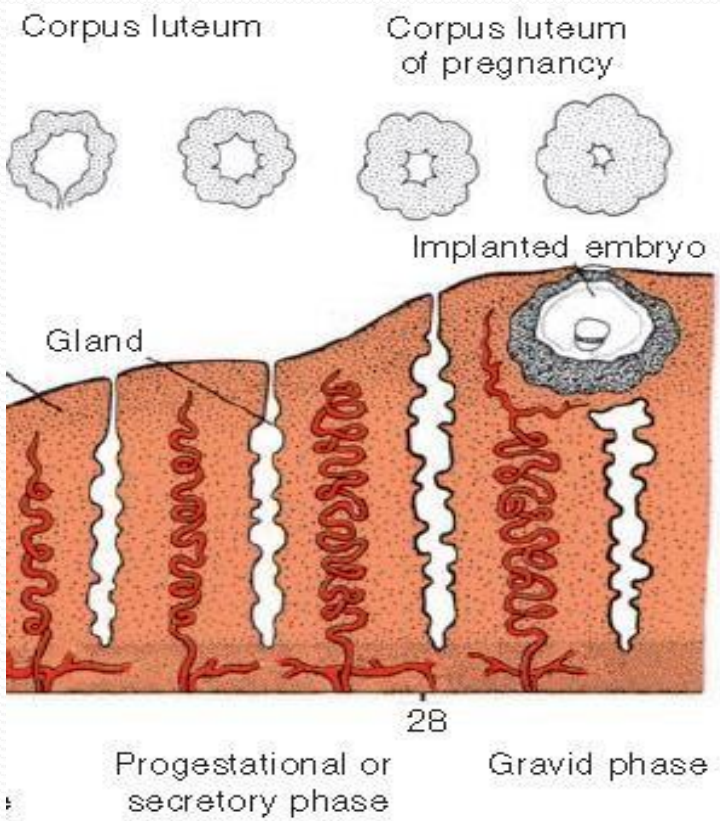


OOCYTE TRANSPORT:

- fimbriae of the oviduct begin to sweep
- tube itself begins to contract rhythmically
- Motion of cilia on the epithelial lining of the tube

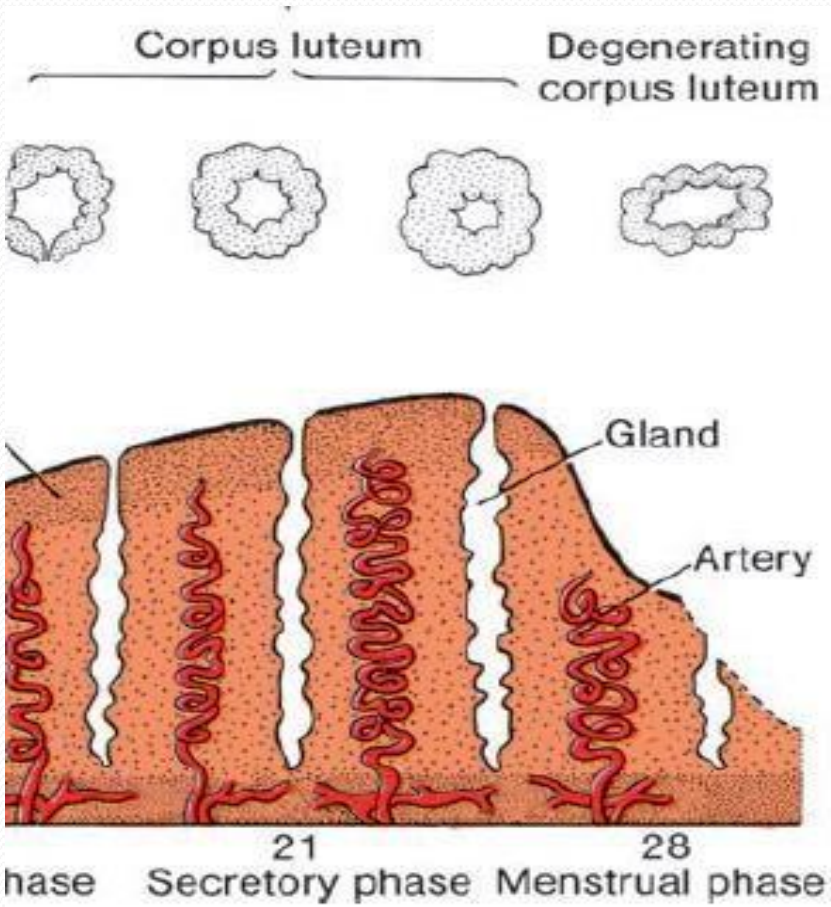


- corpus luteum reaches maximum development approximately 9 days after ovulation
- Its degeneration is prevented by human chorionic gonadotropin (hCG), secreted by the syncytiotrophoblast of the developing embryo
- luteal cells continue to secrete progesterone until the end of the fourth month



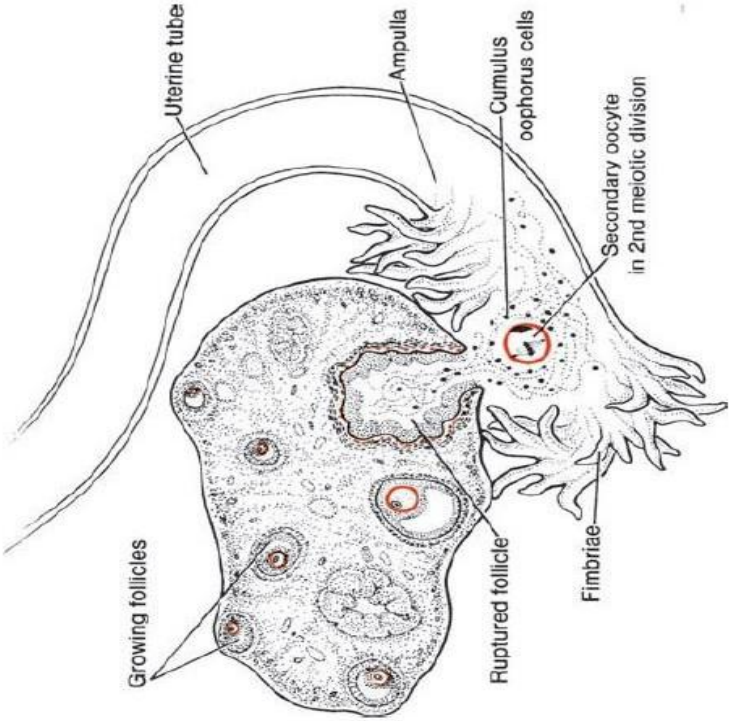
- **If fertilization does not occur:**

- corpus luteum **shrinks** to form the corpus albicans
- progesterone production **decreases**
- precipitating **menstrual bleeding**



Fertilization

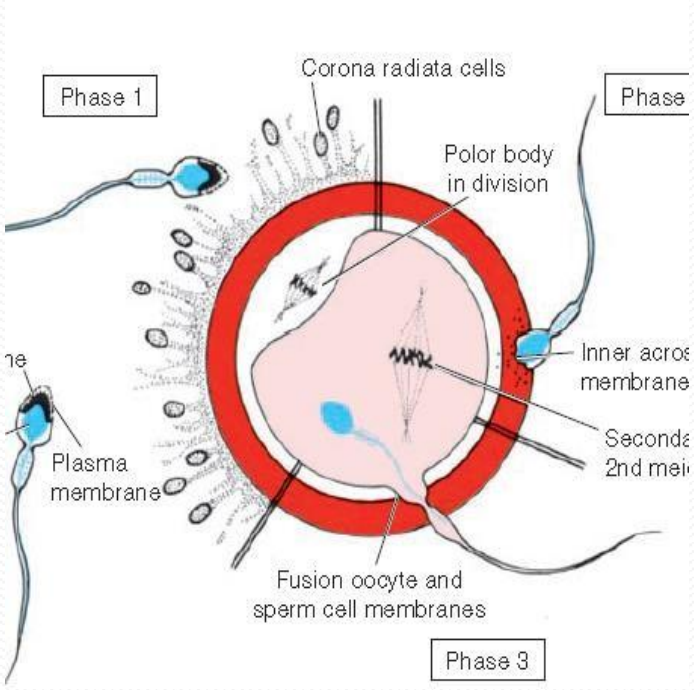
- the process by which male and female gametes fuse
- occurs in the ampullary region of the uterine tube
- The trip of sperms from cervix to oviduct requires a minimum of 2 to 7 hours.



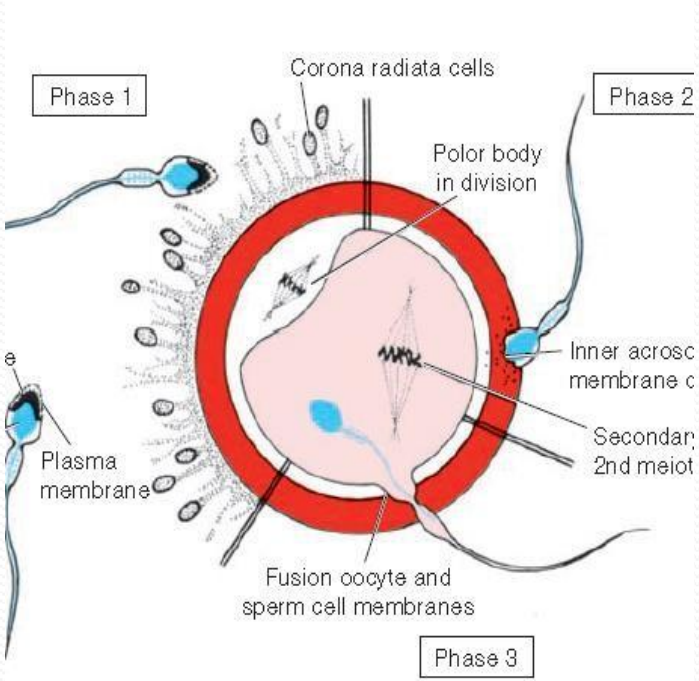
- Spermatozoa are not able to fertilize the oocyte immediately but must undergo
- (*a*) ***capacitation*** : glycoprotein coat and seminal plasma proteins are removed

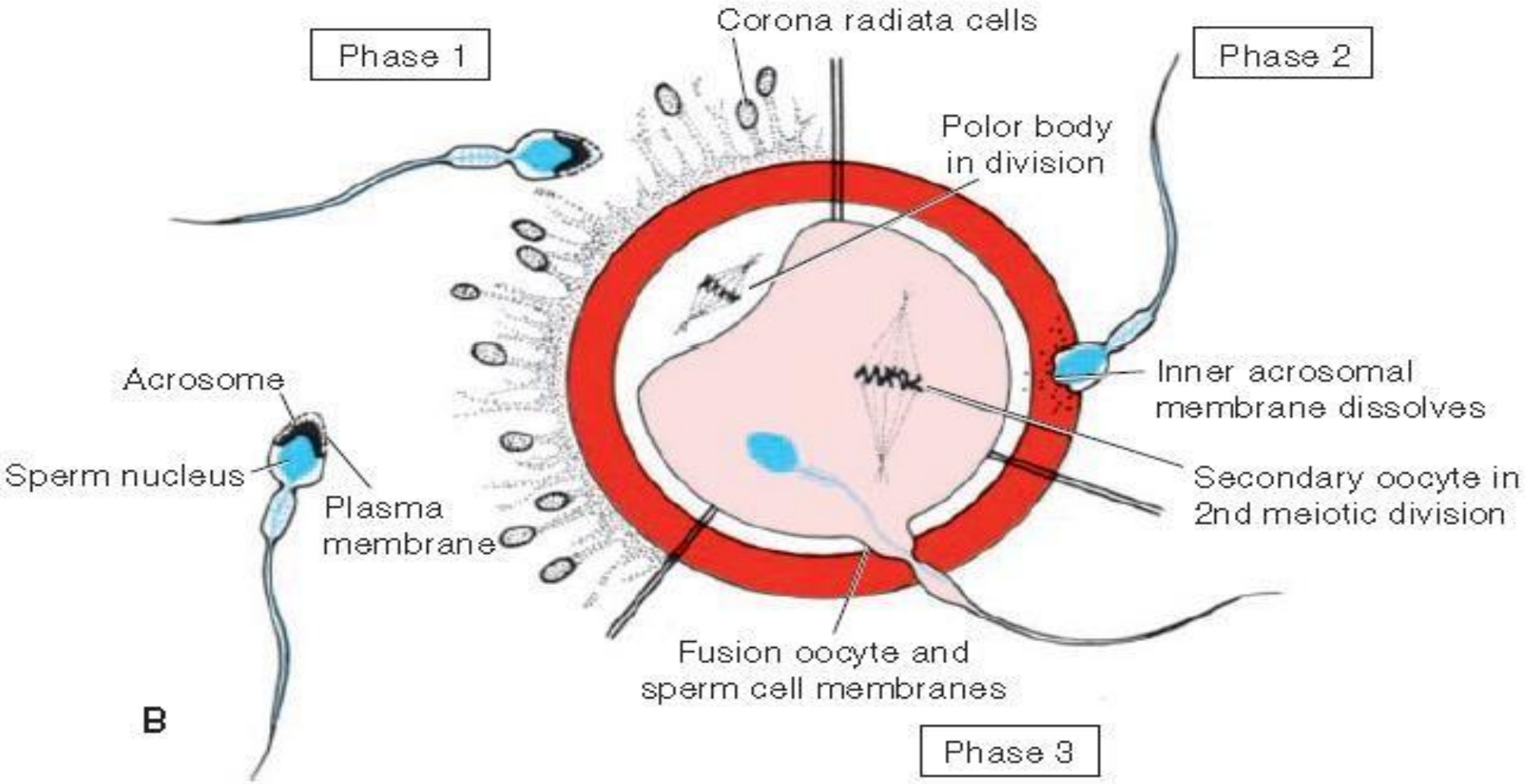
(*b*) ***acrosome reaction***:

- after binding to the zona pellucida
- release of enzymes needed to penetrate the zona pellucida



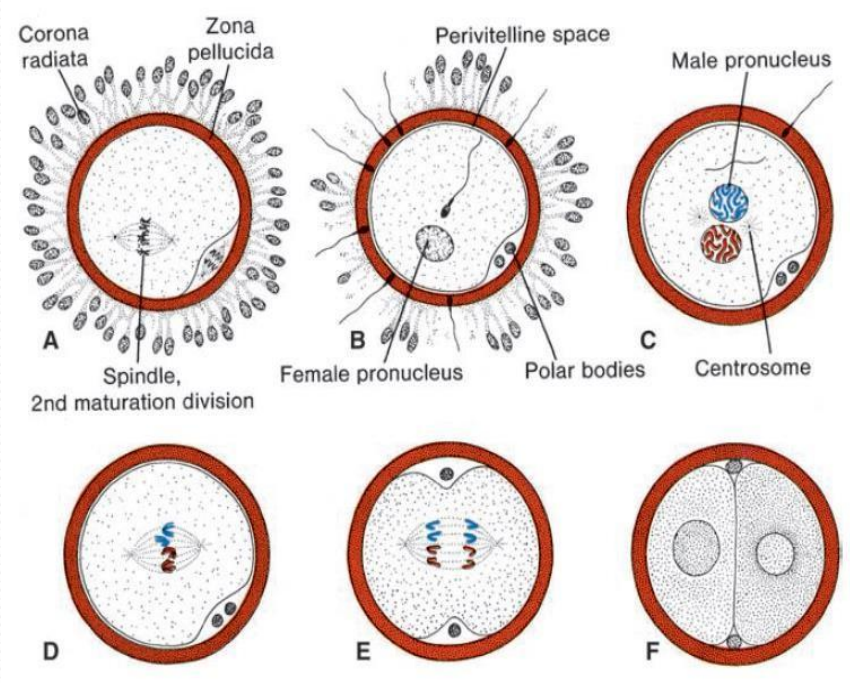
- The phases of fertilization:
- Of the 200 to 300 million spermatozoa deposited in the female genital tract, only 300 to 500 reach the site of fertilization. Only one of these fertilizes the egg
- phase 1, penetration of the corona radiata
- phase 2, penetration of the zona pellucida
- phase 3, fusion of the oocyte and sperm cell membranes.




Phase 1
Corona radiata cells
Phase 2
Polor body
in division
Acrosome
Inner acrosomal
membrane dissolves
Sperm nucleus
Plasma
membrane
Secondary oocyte in
2nd meiotic division
Fusion oocyte and
sperm cell membranes
B
Phase 3

PHASE 3: FUSION OF THE OOCYTE AND SPERM CELL MEMBRANES:

1. **Cortical and zona reactions** :the oocyte membrane becomes impenetrable to other spermatozoa
2. **Resumption of the second meiotic division**
3. **Metabolic activation of the egg**



- **The main results of fertilization are as follows:**

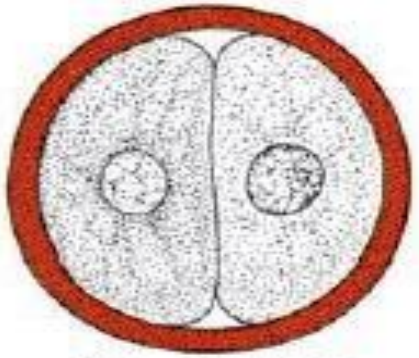
A. **Restoration of the diploid number of chromosomes (fromation of zygote)**

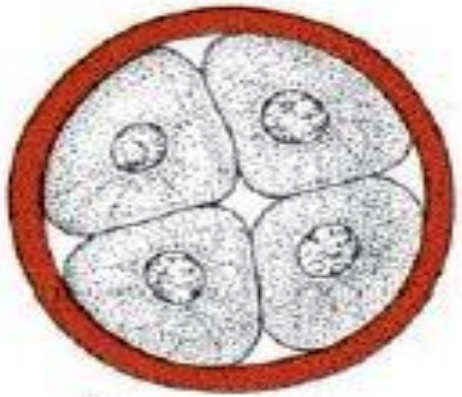
B. **Determination of the sex**

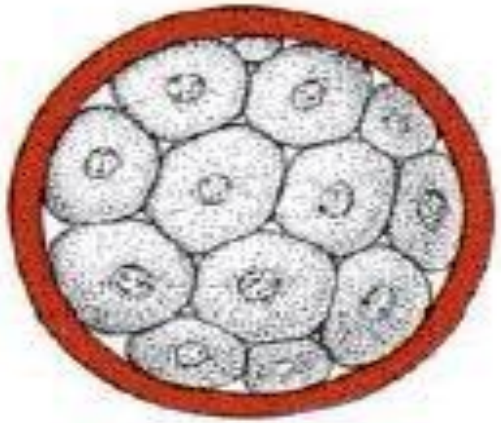
C. **Initiation of cleavage**

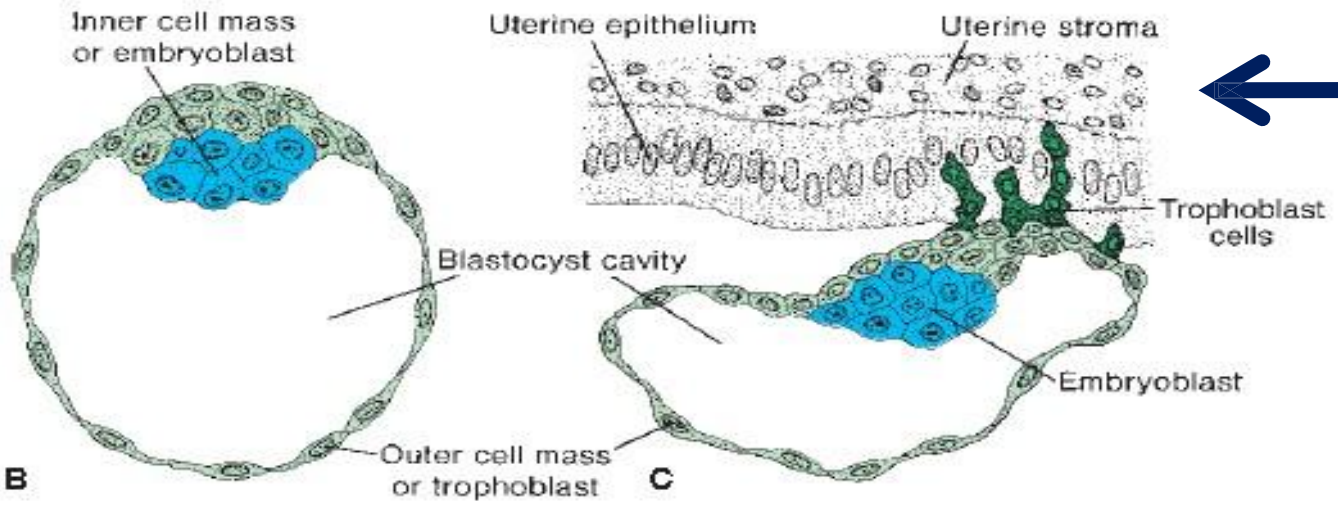
Cleavage

- Zygote
- Blastomeres
- Morula
- Blastocyst
- Uterus at Time of Implantation









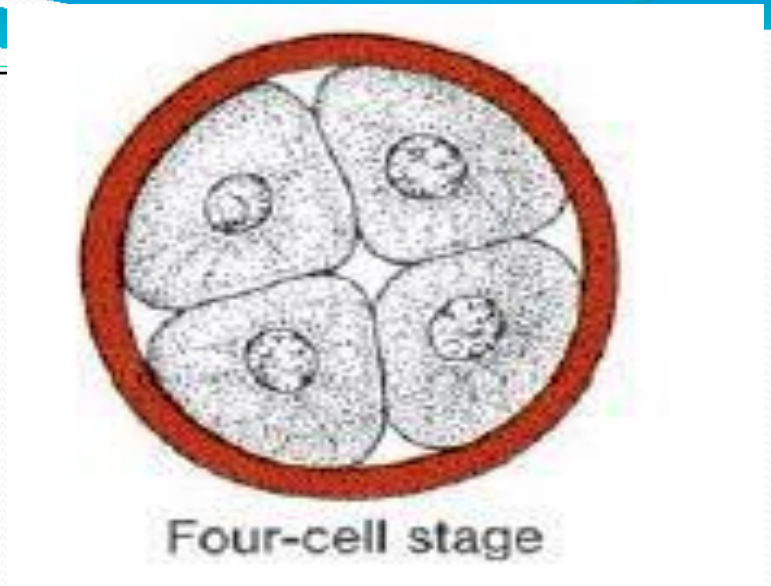
- **Zygote**
  - **As a result of fertilization**
  - **One cell (46 xx or xy)**

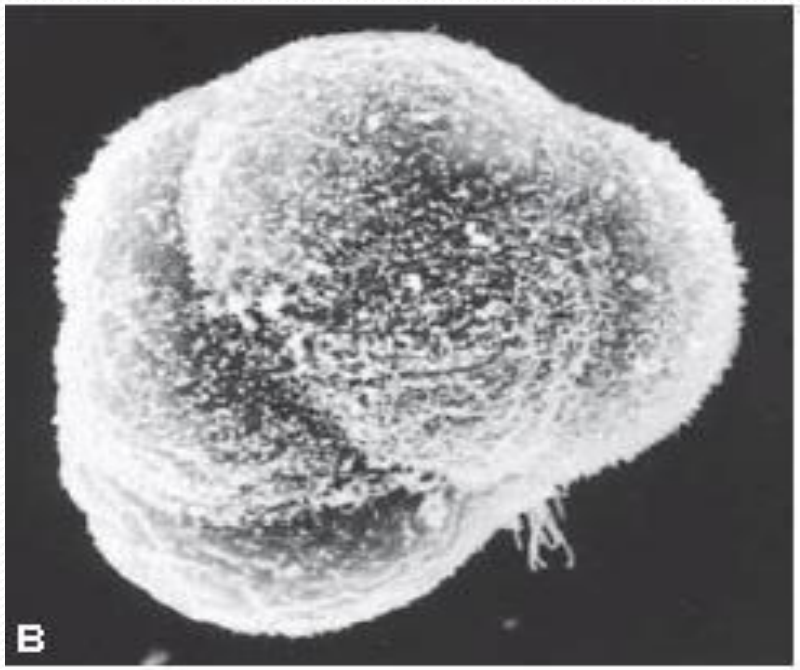
- **Blastomeres**

✓ Four -cell stage

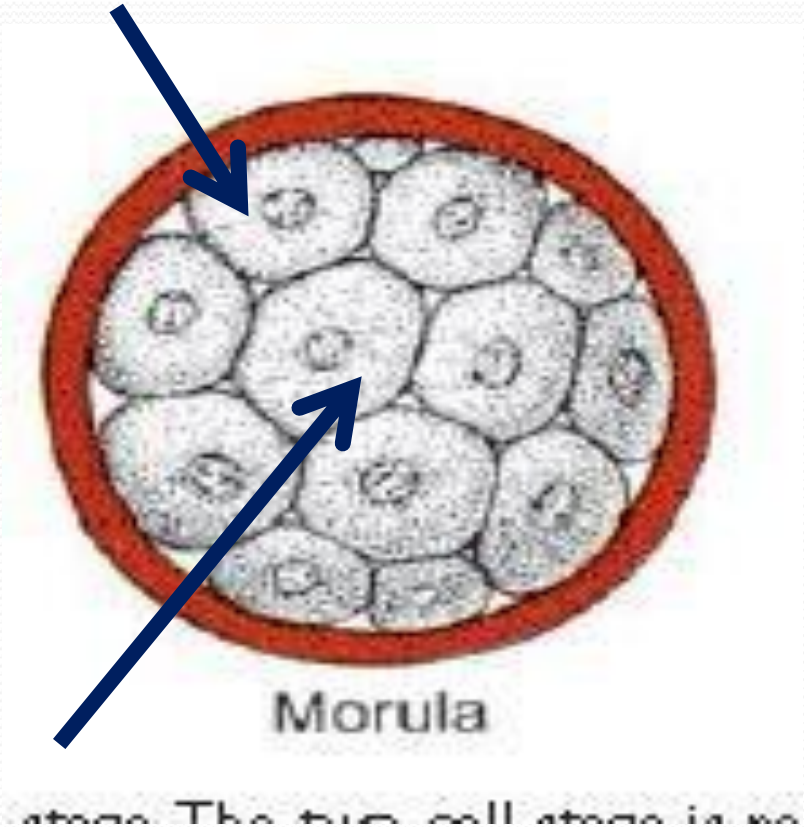
✓ eight -cell stage

✓ Forming a compact ball of cells held together (compaction)

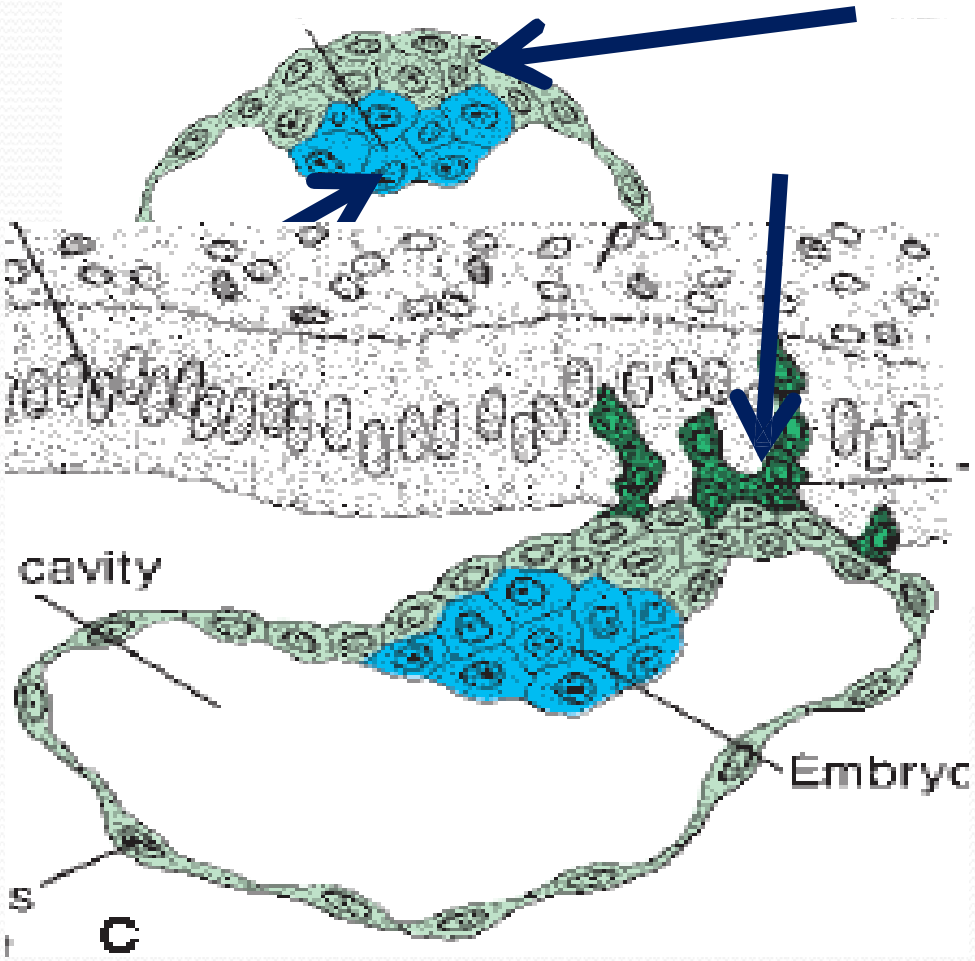




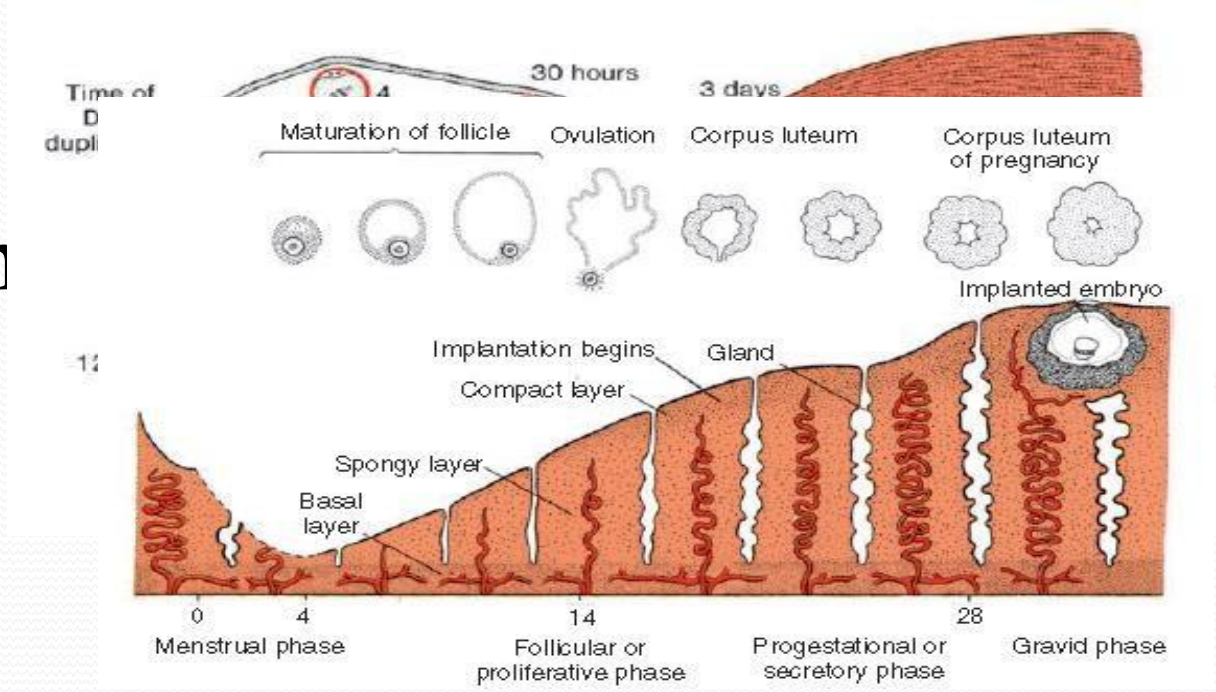
- **morula**
  - 16-cell
  - inner cell mass
    - embryo proper
  - outer cell mass.
    - trophoblast, which later contributes to the placenta.



- **Blastocyst**
  - ✓ single cavity, the blastocele
  - ✓ inner cell mass became embryoblast
  - ✓ outer cell mass (trophoblast)
    - ➢ flatten and form the epithelial wall of the blastocyst
    - ➢ begin to penetrate the uterine mucosa(embryonic pole)



- **Uterus at Time of Implantation:**
- **three distinct layers can be recognized in the endometrium**
  - **compact layer**
  - **spongy layer**
  - **basal layer**



CLINICALCORRELATES

Contraceptive Methods:

1. Barrier techniques
2. contraceptive pill
3. intrauterine device (IUD)
4. Vasectomy and tubal ligation

- **Infertility:**
- problem for 15% to 30% of couples

1. insufficient numbers of sperm and/or poor motility
2. occluded oviducts
3. immunity to spermatozoa
4. absence
5. of ovulation, and others

Thank you…